

Zonal Adaptive Dynamics Operating System

A bio-inspired cognitive architecture that gives LLMs persistent memory, emotional modulation, structured reasoning, and adaptive learning.

THE PROBLEM

Large language models are stateless text generators. They have no persistent memory, no internal emotional state, no structured reasoning pipeline, and no capacity to learn from their own mistakes. Every conversation starts from zero.

THE SOLUTION

ZADOS wraps any LLM in a full cognitive nervous system — simulated neurotransmitters, 29 specialized reasoning engines, 3-tier memory with consolidation, and a reward system that shapes every response. The LLM becomes one component in a much larger cognitive architecture.

HOW IT WORKS — 7-PHASE COGNITIVE PIPELINE

Perception → Neurochemical Modulation → Engine Dispatch → Internal Thinking → Reward Evaluation → Response Generation → Learning

CORE SYSTEMS

Neurochemical Engine	29 Cognitive Engines	3-Tier Memory	Reward System
12 neurotransmitters modeled via stochastic differential equations. Receptor binding, 5 oscillatory bands, cross-frequency coupling. Creates continuous internal state that colors all cognition.	13 clusters: contradiction detection, Socratic reasoning, SOAR production rules, AtomSpace hypergraph, pattern analysis, strategic planning, emotional processing, metacognition, and more.	Short-term (working), mid-term (session), long-term (persistent). Automatic consolidation via emotion-driven promotion. Identity, knowledge, and thought namespaces.	Tonic (sustained) + Phasic (reactive) pathways. Scores on logic, ethics, innovation, attunement. Generates response-shaping directives. 17 contextual profiles.

WHAT MAKES ZADOS DIFFERENT

- **Multi-Pass Reasoning:** Every response passes through internal reasoning, then reward scoring, then generation — not one-shot.
- **Persistent Learning:** Tracks what works across sessions; adjusts meta-parameters autonomously.
- **Sleep & Dreaming:** Dedicated REM and Dream modes for memory consolidation and creative recombination.
- **Continuous Internal State:** 12-NT simulation creates a genuine internal state that evolves over time and modulates cognition.
- **Emotional Grounding:** 46-emotion taxonomy drives real computational changes, not surface decoration.
- **Model-Agnostic:** The entire cognitive layer is LLM-agnostic. The model enters at only 2 of 7 phases.

CURRENT STATUS

6,015+	29	230+	5
Passing Tests	Cognitive Engines	Source Files	Learning Modes

Tech: Python 3.13 · NumPy (SDE integration) · Ollama/LLaMA (local, swappable) · Zero ML training — all engines are algorithmic · pytest